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SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%
Software:Cisco Packet Tracer, Wireshark

Code of Conduct:

I G. Santosh Kumar *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G. Santosh Kumar

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Problem Identification	20%				
Debugging	25%				
Optimization	20%				
Analysis	20%				
Code Quality	15%				

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26.**

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

G. Santosh Kumar
192524020
CSA0707

CO2 - AT3 - Debugging

I. Debugging IPv4 Network Configuration

Given

- IP address : 192.168.10.65
- Subnet Mask: 255.255.255.192 (/26)
- Default Gateway: 192.168.10.1

Step-01 Find the subnet

A /26 subnet has:

- Block size = 64
- Networks :

Network	Host Range	Broadcast
192.168.10.0/26	1-62	63
192.168.10.64/26	65-126	127
192.168.10.128/26	129-190	191
192.168.10.192/26	193-254	255

The IP 192.168.10.65 belongs to

192.168.10.64/26

St:-02 Network Information

Network Address

192.168.10.64

Broadcast Address

192.168.10.127

Valid Host Range

192.168.10.65 – 192.168.10.126

St:-03 Verify IP address

Configured IP

192.168.10.65

• Valid (first usable Host)

St:-04 Verify Gateway

Gateway Configured

192.168.10.1

Gateway belongs to subnet

192.168.10.0/26

The workstation belongs to

192.168.10.64/26

Gateway is in a different subnet.

St:-05 Configuration Errors

- IP is correct
- Gateway is incorrect
- Devices in 192.168.10.64/26 cannot use gateway 192.168.10.1.

St:-06 Correct Configuration

Example:-

IP address

192.168.10.65

Gateway

St:-07 Number of Usable Hosts

Formula

$$2^{(32-26)}-2$$

$$2^{6}-2 = 64-2=62$$

2. IPv6 Debugging

Device A

2001:db8::ff00:42:8329/64

Expanded

2001:0db8:0000:0000:0000:ff00:0042:8329

Device B

2001:db8:0:0:1::1/64

Expanded

2001:0db8:0000:0000:0001:0000:0000:0001

Network prefix

A /64 means first four blocks.

Device A

2001:0db8:0000:0000

Device B

2001:0db8:0000:0000

Both prefixes are identical

Communication fails?

The IPv6 address are correct.

possible reasons:

- firewall blocking ICMPv6
- wrong default gateway
- Duplicate address.
- Interface disabled.
- switch/router issue.

optimized configuration

Device A

2001:db8:0:0::10/64

Device B

2001:db8:0:0::20/64

Gateway

2001:db8:0:0::1

Available Host address

Host bits

128 − 64 = 64

= 18,446,744,073,709,551,616 addresses

3. IPv4 Subnet optimization (VLSM)

Network

192.168.1.0/24

/26 subnet

Network

192.168.1.0

Broadcast

192.168.1.63

Hosts

192.168.1.1 – 192.168.1.62

Usable

62

/27 Subnet

Network

192.168.1.64

Broadcast

192.168.1.95

Hosts

192.168.1.65 – 192.168.1.94

usable

30

subnet

Network

192.168.1.96

Broadcast

192.168.1.111

Hosts

192.168.1.97 – 192.168.1.110

Usable

14

Address utilization

Subnet	Usable Hosts
/26	62
/27	30
/28	14

9. IPv6 Debugging

Expanded address

Device A

2001:0db8:0000:0000:0000:ff00:0042:8329

Device B

2001:0db8:0000:0000:0001:0000:0000:0001

Network prefix

2001:db8:0:0::/64

optimized

2001:db8:0:0::10/64

2001:db8:0:0::20/64

Gateway:

2001:db8:0:0::1

Hosts

= 18,446,744,073,709,551,616

5. Routing Table Debugging

Routing Table

Destination	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default	10.0.0.1

Destination

192.168.2.100

st prefix match

Destination

192.168.2.100

matches

192.168.2.100

This is the longest prefix.

Next Hop

packet forwarded to

10.0.0.3

Destination verification

Network

192.168.2.0/24

Host Range

192.168.2.1

to

192.168.2.254

192.168.2.100

valid

possible Routing Errors

If packets fail:

- wrong next-hop Ip
- Router at 10.0.0.3 is down.
- Missing return route.
- Interface down.
- Incorrect subnet Mask.
- Access control list (ACL) blocking traffic.

optimized Routing Table

Destination	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default	10.0.0.1

If many subnets share the same path, route summari-zation can reduce the routing table size.

If no matching route exists

The router uses the default route (0.0.0.0/0).

Next Hop: 10.0.0.1